

4.

$$F(s) = \frac{s+3}{(s+1)^2(s+2)} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s+2}$$

兩邊同乘 $(s+1)^2(s+2)$:

$$\begin{aligned} s+3 &= A(s+1)(s+2) + B(s+2) + C(s+1)^2 \\ &= A(s^2+3s+2) + Bs+2B + C(s^2+2s+1) \\ &= (A+C)s^2 + (3A+B+2C)s + (2A+2B+C) \end{aligned}$$

$$\begin{cases} A + C = 0 \\ 3A + B + 2C = 1 \\ 2A + 2B + C = 3 \end{cases}$$

$$C = -A$$

$$A+B=1$$

$$A+2B=3$$

$$B=2, A=-1, C=1$$

$$F(s) = \frac{-1}{s+1} + \frac{2}{(s+1)^2} + \frac{1}{s+2}$$

$$L^{-1} \left\{ \frac{1}{s+a} \right\} = e^{-at}$$

$$L^{-1} \left\{ \frac{1}{s+a^2} \right\} = te^{-at}$$

$$L^{-1} \left\{ \frac{-1}{s+a} \right\} = -e^{-t}$$

$$L^{-1} \left\{ \frac{1}{s+1^2} \right\} = 2te^{-t}$$

$$L^{-1} \left\{ \frac{1}{s+2} \right\} = e^{-2t}$$

$$f(t) = (2t-1)e^{-t} + e^{-2t}$$

6.

求 $f(t)=te^{-5}$ 的傅立葉變換。

解：

$$F(\omega)=\int_0^{\infty} f(t)e^{-j\omega t} dt$$

$$=\int_0^{\infty} t e^{-(5+j\omega)t} dt$$

$$\text{令 } a=5+j\omega$$

則

$$F(\omega)=\int_0^{\infty} t e^{-at} dt$$

利用公式

$$=\int_0^{\infty} t e^{-at} dt = \frac{1}{a^2}$$

因此

$$F(\omega)=\frac{1}{(5+j\omega)^2}$$

答：

$$F(\omega)=\frac{1}{(5+j\omega)^2}$$